João Pedro Gonçalves Ramos

IMPA

April 28th, 2026

My path

2015–2016	UFRJ	Undergraduate
2014–2016	IMPA	Master's
2016–2020	Bonn	PhD
2020–2024	IMPA	
	ETH Zürich	Postdocs
	EPFL	
2024–2025	ULisbon	Tenure-track
2025–current	IMPA	Tenure-track (current)

My work so far

Focus on Analysis in the broad sense:

Harmonic Analysis

Partial Differential Equations

Calculus of Variations

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Highlights

Stability of Concentration for STFT (*Invent. Math.*)

Interpolation formulas and Fourier uniqueness (*JEMS*, *Anal. PDE*, *Ann. PDE*)

Stability for sphere packings in dimensions 8 and 24 (*Crelle*)

The big question

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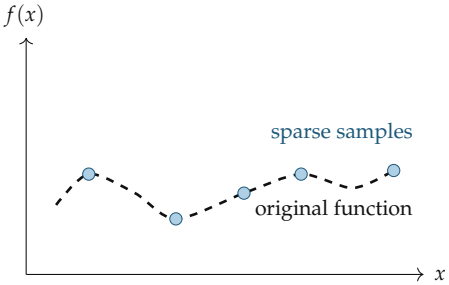

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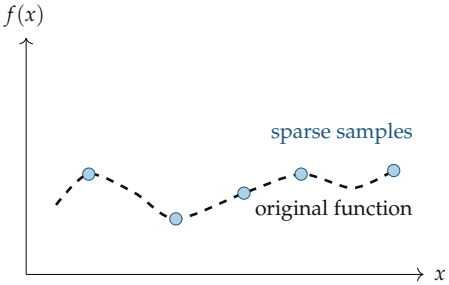
- ∼ We only see sparse samples in space and time.



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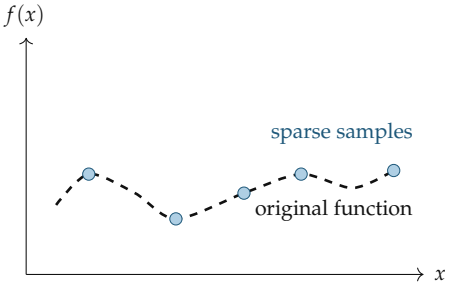

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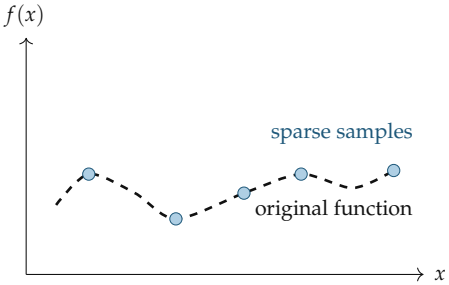

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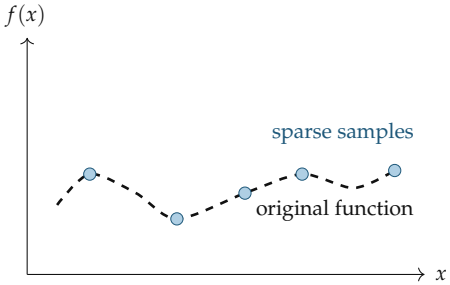

- Classical examples: Heisenberg's U.P., Shannon interpolation.

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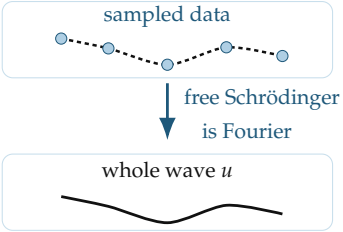

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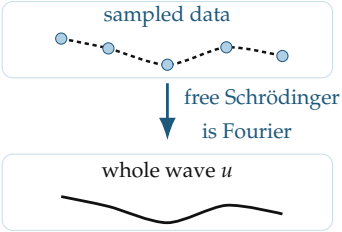

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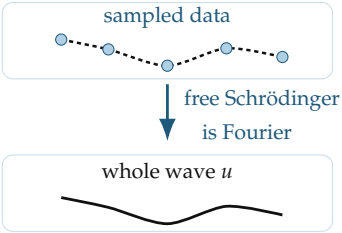

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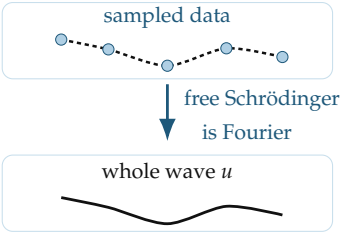

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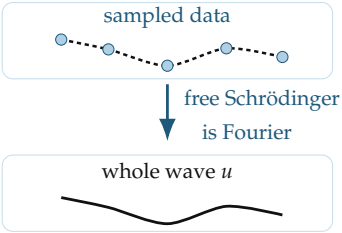

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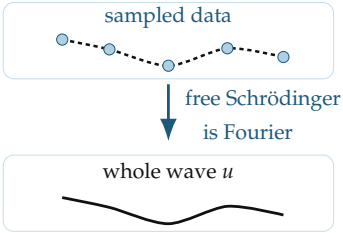

- Ex.: $\boxed{i\partial_t u + \Delta u = |u|^2 \cdot u}$

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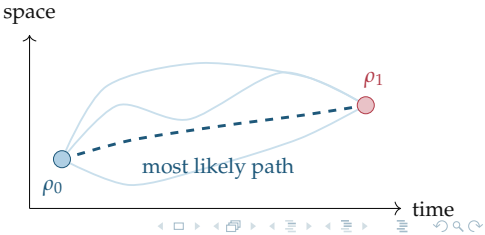

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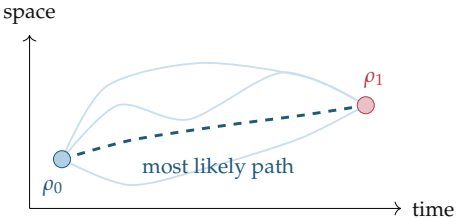

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Discrete recovery of Schrödinger equations

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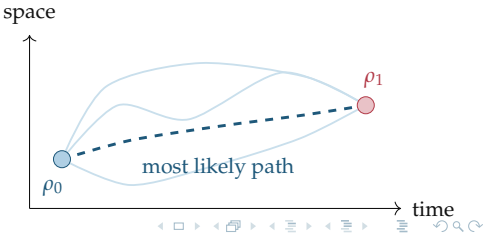
- ∼ Does it hold for discrete measurements?
- ∼ Echoes Schrödinger's own *bridge problem*: from two snapshots, recover the most likely trajectory between them.



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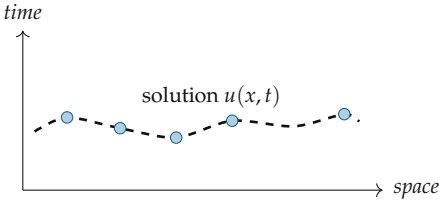
Conjecture: Fixed sparse samples in space and time can determine the whole evolution of quantum systems.



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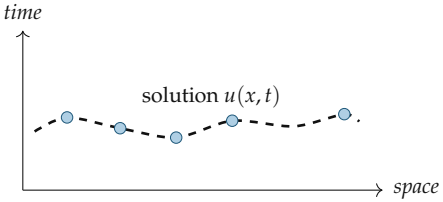

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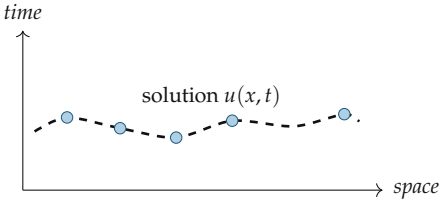

- ∼ Reconstruction from very limited sampled data.
- ∼ First in linear models.
- ∼ Then in nonlinear evolutions.

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- **Bold project:** asks for reconstruction well beyond the Fourier world.
- **Original:** unifies discrete uncertainty and unique continuation in a single framework.
- **Return:** changes how we think about wave phenomena; opens a route toward algorithmic reconstruction; **unlimited potential.**
- **Opportunity:** My work sits exactly on both sides of this bridge: Fourier uniqueness and dispersive PDE.

Very little information may be enough to determine a wave.

Thank you.